

WALNUT RIDGE, AR, USA

WMO: 723406

Lat: 36.125N

Lon: 90.924W

Elev: 83

StdP: 100.33

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93991

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.0	-7.1	-17.0	0.9	-8.3	-13.8	1.1	-5.4	12.2	9.3	11.1	8.7	4.1	0	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	35.1	25.3	33.8	24.9	32.7	24.5	27.0	32.8	26.2	32.1	25.6	31.1	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.7	30.2	24.4	19.5	29.4	23.9	18.9	28.9	85.1	32.9	81.7	32.2	78.9	31.6	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.3	8.3	DB	-12.9	37.6	2.0	1.7	-14.3	38.9	-15.4	39.9	-16.5	40.8	-18.0	42.1	(4)
(5)			WB	-13.6	28.0	2.0	0.9	-15.0	28.6	-16.2	29.1	-17.3	29.6	-18.7	30.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	3.1	5.3	10.2	15.6	20.9	25.6	26.9	26.0	22.1	16.3	9.6	4.6
	DBStd	9.55	5.83	5.68	5.34	4.49	3.99	2.83	2.53	2.60	3.70	4.68	4.96	5.36
	HDD10.0	701	224	151	63	8	1	0	0	0	0	6	67	181
	HDD18.3	2014	472	366	256	103	23	1	0	0	11	96	264	425
	CDD10.0	2736	11	19	70	177	338	467	523	497	363	201	54	14
	CDD18.3	1010	0	0	4	22	102	217	265	239	124	33	2	1
	CDH23.3	9721	0	2	25	171	819	2212	2787	2316	1068	303	17	1
CDH26.7	4024	0	0	1	29	256	969	1258	1022	408	78	2	0	
Wind	WSAvg	3.7	4.3	4.4	4.6	4.8	3.9	3.4	2.9	2.7	2.8	3.3	3.9	4.0
Precipitation	PrecAvg	1200	88	87	110	122	120	79	97	79	88	93	114	109
	PrecMax	1607	182	208	281	281	201	166	188	199	210	291	274	229
	PrecMin	894	26	28	37	20	21	36	25	6	2	0	25	7
	PrecStd	183	40	44	50	55	53	33	48	42	47	59	72	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	22.4	26.2	29.1	32.9	36.2	37.3	37.0	34.2	31.3	26.2	20.3
		MCWB	17.1	17.5	18.8	20.0	22.6	24.9	25.6	26.2	23.8	22.4	18.5	16.8
	2%	DB	17.4	18.9	23.5	27.5	31.3	34.0	35.2	34.8	32.7	28.8	22.3	17.6
		MCWB	15.1	15.3	17.0	19.2	22.3	24.2	25.7	25.7	23.7	20.8	16.4	15.2
	5%	DB	14.9	17.2	21.2	25.9	29.1	32.9	33.8	33.0	31.2	27.1	19.8	15.3
		MCWB	12.1	13.3	15.8	18.2	21.6	23.8	25.5	25.1	23.1	19.0	15.9	12.6
	10%	DB	12.3	14.1	18.8	23.2	27.7	32.1	32.6	32.1	28.9	24.1	17.5	12.7
		MCWB	9.7	10.9	14.5	16.9	20.7	23.7	25.3	24.8	21.8	17.7	13.7	10.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	18.4	19.9	22.1	24.9	27.0	27.9	27.9	25.8	24.1	20.1	17.7
		MCDB	18.7	21.3	24.6	27.1	30.0	32.8	33.5	33.7	31.9	29.2	23.4	19.5
	2%	WB	15.7	16.0	18.4	20.6	23.8	25.8	27.0	27.0	25.0	22.0	17.9	15.8
		MCDB	17.2	18.5	22.5	25.5	28.6	31.8	32.8	32.8	30.8	26.7	20.6	17.5
	5%	WB	12.8	14.1	16.8	19.4	22.8	25.1	26.3	26.2	24.1	20.7	16.3	13.2
		MCDB	14.2	16.5	20.0	23.9	27.8	31.0	32.1	32.0	29.2	24.6	19.2	14.9
	10%	WB	9.8	11.4	15.0	18.1	21.8	24.4	25.7	25.5	23.2	19.2	14.4	10.9
		MCDB	12.3	14.2	18.1	22.2	26.6	30.0	31.2	30.6	27.3	23.1	17.3	12.7
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.4	10.5	11.1	10.3	10.3	9.7	10.4	11.5	12.2	10.1	8.4
		MCDBR	11.5	13.3	13.1	13.0	12.0	11.5	11.0	11.8	13.1	14.3	13.4	10.9
		MCWBR	9.5	10.0	8.3	7.1	5.6	4.4	4.1	5.2	5.8	7.3	8.9	8.9
	5% WB	MCDBR	9.9	11.4	11.3	10.6	10.2	10.2	10.2	10.8	11.1	11.9	11.2	9.7
		MCWBR	9.3	9.9	8.1	6.9	5.5	4.6	4.4	5.2	5.4	6.7	8.7	9.0
Clear-Sky Solar Irradiance	taub	0.316	0.322	0.357	0.401	0.437	0.459	0.486	0.476	0.413	0.355	0.324	0.310	
	taud	2.476	2.458	2.392	2.268	2.229	2.181	2.146	2.153	2.322	2.469	2.505	2.510	
	Ebn at Noon	881	918	910	883	852	830	803	804	841	869	865	866	
	Edh at Noon	84	96	112	133	141	148	152	148	119	94	81	76	
All-Sky Solar Radiation	RadAvg	2.38	2.95	4.06	5.17	5.74	6.55	6.29	5.74	4.87	3.70	2.60	2.03	
	RadStd	0.21	0.38	0.32	0.35	0.45	0.42	0.30	0.35	0.34	0.43	0.28	0.22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page